

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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2025-11-11

Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

# Load the MA_schools dataset
data(MA_schools)
```

Data Exploration (3 points)

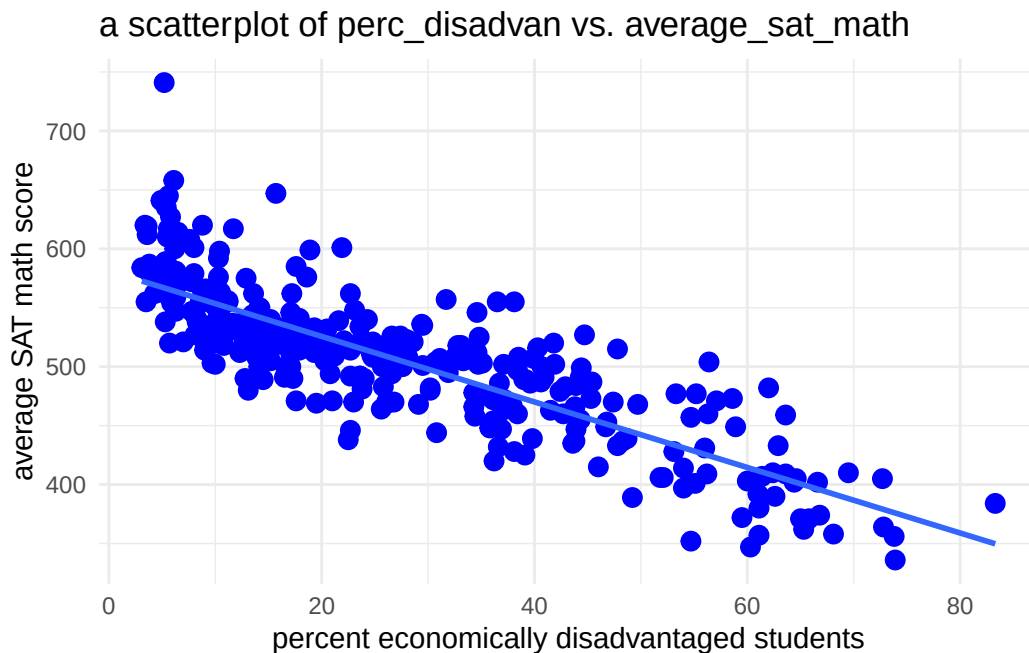
1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
# YOUR CODE HERE
```

```
ggplot(MA_schools, aes(x=perc_disadvan,y=average_sat_math)) +  
  geom_point(size = 3, color = "blue") +  
  geom_smooth(method= "lm", se = FALSE) +  
  labs(title = "a scatterplot of perc_disadvan vs. average_sat_math",  
        y = "average SAT math score",  
        x = "percent economically disadvantaged students") +  
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Answer: Based on the plot, I expect a negative relationship between percent economically disadvantaged students and average SAT math score. Schools with a higher percentage of

economically disadvantaged students tend to have lower average SAT math scores, and vice versa.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
# YOUR CODE HERE

model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)
summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
# YOUR CODE HERE
```

```
sat_effects <- coef(model1)
sat_effects
```

```
(Intercept) perc_disadvan
581.281054      -2.779787
```

Intercept: 581: When there are 0% economically disadvantaged students, the model predicts an average SAT math score of about 581.

Slope: -2.78: The slope of -2.78 indicates that for each 1% point increase in economically disadvantaged students, the predicted average SAT math score decreases by about 2.78 points.

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: The slope coefficient is -2.78, which represent a negative relationship between the percentage of economically disadvantaged students and average SAT math scores. Schools with higher percentages of economically disadvantaged students tend to have lower average SAT math scores. For each 1% increase in economically disadvantaged students, the average SAT math score drops by about 2.78 points

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
# YOUR CODE HERE
```

```
new_data = data.frame(perc_disadvan = 50)
predict(model1, newdata = new_data)
```

```
1
442.2917
```

Predicted average SAT math score: At a school where 50% of students are economically disadvantaged, the model predicts an average SAT math score of about 442.

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
# YOUR CODE HERE
```

```
predict(model1, newdata = new_data, interval = "prediction")
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
# YOUR CODE HERE
```

```
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

```
      Min      1Q  Median      3Q      Max
-81.342 -20.802  -3.872  18.211 167.251
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   588.1898     7.6067  77.325  <2e-16 ***
perc_disadvan  -2.7772     0.1063 -26.120  <2e-16 ***
```

```

sizemedium    -11.9134      7.5346   -1.581    0.115
sizelarge     -6.3615      6.9231   -0.919    0.359
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom
Multiple R-squared:  0.6988,    Adjusted R-squared:  0.696
F-statistic: 253.6 on 3 and 328 DF,  p-value: < 2.2e-16

```

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```

# YOUR CODE HERE

summary(model1)$r.squared

```

```
[1] 0.6961948
```

```
summary(model2)$r.squared
```

```
[1] 0.6987587
```

Answer: Model 2 (R-squared 0.698) explains slightly more variance in average SAT math scores than Model 1 (R-squared 0.696).

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```

# YOUR CODE HERE

anova(model1, model2)

```

Analysis of Variance Table

Model 1: average_sat_math ~ perc_disadvan

Model 2: average_sat_math ~ perc_disadvan + size

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	330	371191				
2	328	368058	2	3132.6	1.3958	0.2491

Conclusion: Based on the ANOVA test ($p = 0.249 > 0.05$), adding size does not significantly improve the model. This means Model 2 is not statistically better than Model 1 at the $\alpha = 0.05$ level.